TEST REPORT FIRES-FR-180-16-AUNE

Specimen no. 1: Partition made of steel frame glazed by PYRAN®
Specimen no. 2: Steel frame “vrame form 50” glazed by PYRAN®

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Reg. No. 041/S-159

The Experts on Fire Safety

TEST REPORT

FIRES-FR-180-16-AUNE

Tested property:	Fire resistance
Test method:	EN 1364-1: 2015
Type of test:	Accredited and Notified (NB 1396)
Date of issue:	20. 10. 2016
Name of the product:	Specimen no. 1: Partition made of steel frame glazed by PYRAN® Specimen no. 2: Steel frame “vrame form 50” glazed by PYRAN®
Manufacturer:	SCHOTT Technical Glass Solutions GmbH, Otto-Schott-Strasse 13, 077 45 Jena, Germany (producer of glass panes) voestalpine Krems GmbH, Schmidhüttenstr. 5, Postfach 43, 3500 Krems, Austria (producer of frame)
Sponsor:	SCHOTT Technical Glass Solutions GmbH, Otto-Schott-Strasse 13, 077 45 Jena, Germany
Test carried out by:	FIRES, s.r.o., Testing laboratory
Task No.:	PR-16-0192
Specimen received:	19. 08. 2016
Date of the test:	23. 08. 2016

Technician responsible for the technical side of this report: Bc. Dávid Šubert

Number of pages: 19
Test reports: 5 Copy No.: 2

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1. INTRODUCTION

This test report contains the results of test carried out by FIRES, s.r.o., Testing laboratory in Batizovce, accredited by SNAS for testing. Certificate of accreditation No.: S-159. The purpose of the test was to gain information for product classification.

Sponsor's representatives witnessing the test:

Ing. Gerrit Panzner	SCHOTT Technical Glass Solutions GmbH
test directed by	Ing. Marek Gorlický
test carried out by	Bc. Dávid Šubert
operator	Miroslav Hudák

2. MEASURING EQUIPMENT

Identification number	Measuring equipment	Note
F 90 020	Vertical test furnace (5 x 5) m for fire resistance testing	-
F 69 010	PLC system for data acquisition and control TECOMAT TC 700	-
F 40 019	Visual and calculating software to PLC TECOMAT TC 700	-
F 40 017	Control and communication software to PLC TECOMAT TC 700	-
F 40 018	SW Reliance	-
F 40 020	Driver Tecomat - Reliance (SW)	-
F 71 043, F 71 044	Transducer of differential pressure (–50 to + 150) Pa	measurement of pressure inside the test furnace
F 54 020	Digital calliper (0 to 200) mm	-
F 54 056	Racking meter	-
F 69 009	PLC system for data acquisition and climate control TECOMAT TC 604	-
F 60 001 - F 60 009	Sensors of temperature and relative air humidity	measurement of climatic conditions
F 13 531 - F 13 548	Plate thermometers	measurement of temperature inside the test furnace according to EN 1363-1
F 13 701	Sheathed thermocouple type K Ø 3 mm	measurement of ambient temperature
F 74 007 - F 74 008	Cable position transducers (0 to 1250) mm	measurement of deflection
F 90 005	Gap gauge for fire resistance testing ∅ 25 mm	-
F 90 006	Gap gauge for fire resistance testing ∅ 6 mm	-
F 57 005	Digital stop-watch	-

3. PREPARATION OF THE SPECIMENS

Testing laboratory noted down production data of specimens from certified production.
Test specimen data are listed in following table:

Frames:

Place of production	voestalpine Krems GmbH, Schmidhüttenstr. 5, Postfach 43, 3500 Krems, Austria
Production number	sample
Date of production	04.08.2016
Date of check out	10.08.2016
Number of certificate ISO 9001	VNA0005058/A
Issuing Body	Loyd's Register EMEA Niederlassung Wien
Date of issuing	21.07.2015

Glass panes:

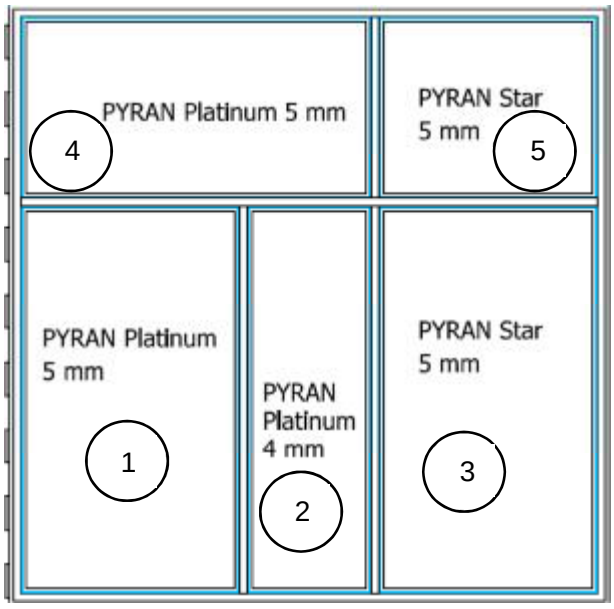
Place of production	SCHOTT Technical Glass Solutions GmbH, Otto-Schott-Strasse 13, 077 45 Jena, Germany
Production number	SAP-Nr: 20041639
Date of production	Cw 32/2016
Date of check out	Cw 33/2016
Number of certificate ISO 9001	36985
Issuing Body	SQS; Bernstrasse 103; CH-3052…
Date of issuing	16.03.2010

Individual parts of test specimens (frames, glass panes, etc.) were delivered to the testing laboratory by test sponsor. Installation of frames to the supporting construction and glazing of glass panes was carried out by workers of the sponsor.

4. PREPARATION OF THE TEST

4.1 DESCRIPTION OF THE SPECIMENS STRUCTURE

Specimen no. 1: Partition made of steel frame glazed by PYRAN®





Dimensions

Overall specimen dimensions (3000 x 3000 x 60) mm (width x height x thickness)

Overall dimensions of glass panes (width x height x thickness):

PYRAN® Platinum	(1097 x 1951 x 5) mm
	(616 x 1951 x 4) mm
	(1763 x 909 x 5) mm
PYRAN® Star	(1097 x 1951 x 5) mm
	(1097 x 909 x 5) mm

Frame

Frame is made of mild steel rectangular profiles with dimensions (60 x 40 x 2) mm, (25 x 25) mm and (35 x 25) mm (position of profiles acc. to drawings). Individual profiles are welded to each other.

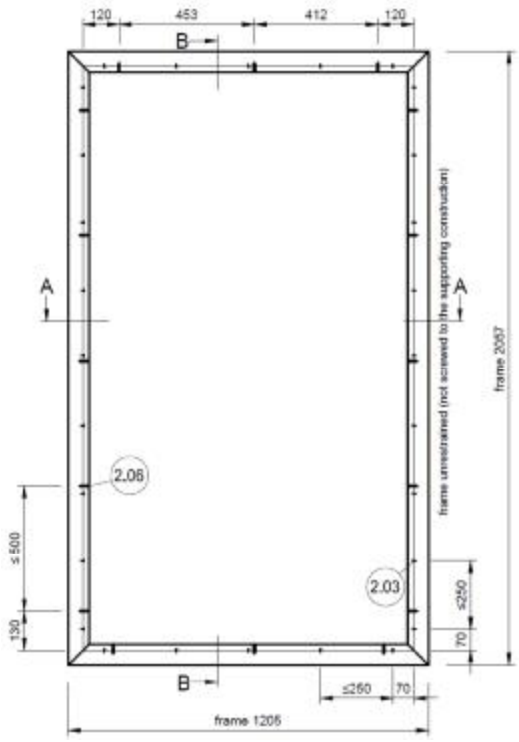
Glazing of glass panes

Glass panes are fixed by glazing beads made of mild steel rectangular profiles (25 x 25 x 2) mm fixed to frame by screws (Ø 4,8 x 45) mm placed in spacing of ≤ 150 mm. (see page 14 for position of screws).

Glazing beads with dimensions (25 x 35 x 2) mm fixed the upper edge of glass panes No. 1 - 3 and are fixed to frame by screws (Ø 4,8 x 45) placed in spacing of ≤ 150 mm. (see page 14 for position of screws).

Glass pane no. 1 and 3 - 5: Gasket Kerafix® Flexlit, 2,0 mm thick (producer: Rolf Kuhn GmbH) is placed between frame/glazing beads and glass pane around the perimeter – two layers on exposed side and one layer on unexposed side.
Glass pane no. 2: One layer of gasket Kerafix® Flexlit, 5,0 mm thick on exposed side and 2,0 mm thick on unexposed side.

Specimen no. 2: Steel frame “vrame form 50” glazed by PYRAN® Platinum, 5,0 mm thick





Dimensions

Overall specimen dimensions (1205 x 2057 x 50) mm (width x height x thickness)
Overall dimensions of glass pane (1097 x 1975 x 5) mm (width x height x thickness)

Frame

Frame is made of galvanized mild steel rectangular profiles, type 5050 L (producer: voestalpine Krems GmbH) (1.01) Individual profiles are welded to each other.

Glazing of glass pane
Glass pane is fixed by glazing beads, type GL 35-8 (2.02) fixed to frame by studs BN65 (2.03) placed in spacing of ≤ 250 mm and steel self-drilling screws (Ø 3,9 x 32) mm (2.06) placed in spacing of ≤ 500 mm. (see page 14 for position of screws).

Gasket, type EVB with dimensions (2 x 15) mm (supplier: voestalpine Krems GmbH) (2.04, 2.05) is placed between frame/glazing beads and glass pane around the perimeter – two layers on exposed side and one layer on unexposed side.

More detailed information about construction of specimens is shown in the drawings which form an integral part of this test report. Drawings were delivered by sponsor.

All the information about technical specifications of used materials and semi-products, information about their type sign were delivered by sponsor. This information was not subject of the specimens verification. Parameters which were verified are quoted in paragraph 4.3.

4.2 DESCRIPTION OF THE SPECIMENS FIXATION

Frames are fixed inside the prepared opening in the supporting construction made of aerated concrete blocks, 250 mm thick with bulk density of 613kg.m^{-3}. Gap between frames and supporting construction are filled with ceramic wool.

Specimen no. 1: Frame is fixed to supporting construction by screws HILTI HUS (Ø 6 x 100) mm placed in spacing of 525 mm on vertical edges and in spacing of 650 mm on horizontal edges. All specimen edges are restrained (fixed). (see page 14 for position of screws).

Specimen no. 2: Frame is fixed to supporting construction by screws HILTI HUS (Ø 6 x 100) mm placed in spacing of 600 mm on one vertical edge and in spacing of ≤ 460 mm on both horizontal edges. Gap between one vertical specimen edge and supporting construction is filled with mineral wool and covered by steel L-profile (30 x 30 x 3) mm fixed to supporting construction only on exposed side. Frame is not directly fixed to the supporting construction on this edge. (see details on page 18)

Orientation of the specimens during the test

Specimen No. 1	glazing beads placed on unexposed side
Specimen No. 2	glazing beads placed on exposed side

4.3 VERIFICATION OF THE SPECIMENS

The conformity of the drawings and the test specimens was verified before and after the fire resistance test. It was possible to carry out this verification on the test specimen. The specimens corresponded to the drawings which are part of this test report. The visual review of the test specimens, the used materials as well as the size verification (basic dimensions) and also the way of specimens fixation to supporting construction were subject of this verification.

4.4 CLIMATIC CONDITIONING OF THE SPECIMENS

Test specimens were stored in the hall of testing laboratory and were conditioned according to EN 1363-1 under the following climatic conditions:

Ambient air temperature [°C]

mean	23,5
standard deviation	2,4

Relative air humidity [%]

mean	50,6
standard deviation	3,5

The humidity equilibrium state of test specimens was not determined. Test specimens did not comprise hygroscopic materials.

5. CARRYING OUT OF THE TEST

5.1 CONDITIONS OF THE TEST

Conditions in the test furnace (temperature – standard temperature/time curve, pressure, content of O_2) as well as in the testing room (ambient temperature) corresponded to EN 1363-1 during the test. Detailed information is part of this test report, or in record from the test.

Values characterizing environment in the testing room directly before the test:

Relative air humidity [%]	Ambient air temperature [°C]
49,2	20,2

5.2 RESULTS OF THE TEST

Measured values are stated in this test report. Description of the specimens behaviour during the test:

Time [min:s]	Face of specimen	Observation
05:00	NS	specimen no. 1: deflection of the frame towards the furnace;
60:00	NS	no further significant visible changes;
180:00	NS	no further significant visible changes;
240:00	NS	no further significant visible changes;
241:15	termination of the test.	

ES: exposed face of specimen; NS: unexposed face of specimen

6. CLOSING

Evaluation of the test:

Specimen no. 1:

Performance criterion	Time till the performance criterion is achieved
integrity – sustained flaming	241 minutes no failure
integrity – gap gauges ∅ 6 mm and ∅ 25 mm	241 minutes no failure
integrity – cotton pad	241 minutes no failure
insulation – average temperature (140 K)	not evaluated
insulation – maximal temperature (180 K)	not evaluated
radiation 15 $kW.m^{-2}$	not evaluated

Specimen no. 2:

Performance criterion	Time till the performance criterion is achieved
integrity – sustained flaming	241 minutes no failure
integrity – gap gauges ∅ 6 mm and ∅ 25 mm	241 minutes no failure
integrity – cotton pad	241 minutes no failure
insulation – average temperature (140 K)	not evaluated
insulation – maximal temperature (180 K)	not evaluated
radiation 15 $kW.m^{-2}$	not evaluated

The fire test was terminated in the 242nd minute upon request of test sponsor.

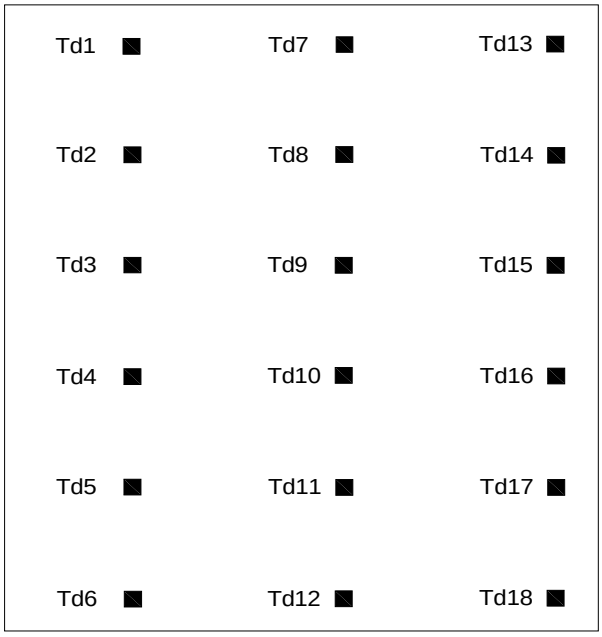
7. DIRECT APPLICATION OF TEST RESULTS

Direct field of application is valid in accordance with Annex A of EN 1364-1: 2015. Validity of individual items of field of direct application shall be determined in classification process.

Measured values inside the test furnace – part 1

Time	Temperature [°C]											
t [min]	Td1	Td2	Td3	Td4	Td5	Td6	Td7	Td8	Td9	Td10	Td11	Td12
0	22,8	24,5	25,3	26,3	28,0	21,6	22,1	19,1	23,8	24,3	24,6	23,4
10	682,3	647,9	641,9	720,7	684,2	678,4	661,8	689,4	719,3	672,2	672,5	680,5
20	774,4	736,0	721,3	763,2	789,1	778,6	745,0	798,7	785,4	763,6	755,5	764,9
30	833,9	800,1	810,1	845,6	820,8	834,4	816,1	833,2	846,7	820,3	821,2	831,3
40	900,0	885,7	859,7	885,2	862,8	895,4	873,5	887,9	901,8	870,9	877,0	889,2
50	912,0	882,7	874,6	847,3	852,2	884,2	867,2	871,0	881,9	845,9	868,0	888,2
60	976,1	943,1	934,4	927,4	957,8	948,6	936,5	945,0	942,3	912,4	941,3	950,4
70	984,3	956,2	946,9	933,4	964,1	958,4	946,5	952,8	950,3	919,7	950,9	960,0
80	1007,0	976,6	970,1	963,1	992,9	983,2	970,6	981,4	973,1	947,4	976,1	983,3
90	1025,7	996,1	989,0	981,1	1011,6	1000,4	988,5	996,9	989,1	963,0	994,4	1001,1
100	1037,2	1005,5	992,3	987,2	1016,4	1010,4	997,8	1009,5	996,8	974,3	1001,6	1009,2
110	1058,1	1026,7	1011,7	1009,4	1032,2	1036,7	1021,1	1036,3	1016,5	992,1	1023,0	1030,8
120	1076,3	1039,3	1018,5	1022,6	1041,1	1051,8	1034,8	1054,1	1030,5	1009,7	1034,7	1044,3
130	1072,8	1033,9	1009,7	1013,4	1032,5	1046,8	1028,4	1051,3	1024,8	1005,0	1027,5	1038,5
140	1075,3	1036,9	1013,4	1015,0	1032,8	1049,3	1031,6	1054,6	1027,0	1008,9	1029,8	1041,3
150	1079,1	1040,3	1016,6	1020,9	1036,3	1053,0	1034,1	1057,8	1029,0	1012,0	1033,5	1044,4
160	1137,7	1103,2	1082,9	1044,3	1020,1	1112,2	1084,5	1084,2	1111,7	1074,5	1074,5	1111,1
170	1149,6	1117,4	1097,2	1059,1	1046,8	1127,4	1104,2	1100,1	1135,6	1097,3	1092,0	1127,5
180	1152,2	1117,3	1095,6	1068,0	1060,3	1131,4	1108,9	1107,0	1142,1	1107,1	1096,9	1130,3
190	1155,3	1123,1	1104,6	1071,0	1062,2	1136,4	1116,3	1113,2	1146,9	1112,5	1102,3	1135,8
200	1157,0	1119,3	1101,8	1072,0	1065,8	1127,3	1124,0	1158,4	1127,9	1163,5	1101,7	1128,5
210	1166,2	1137,3	1105,7	1077,5	1066,7	1126,4	1121,6	1139,5	1117,0	1161,8	1105,9	1128,8
220	1159,8	1124,9	1105,4	1071,7	1060,1	1129,8	1122,6	1144,6	1125,2	1160,0	1102,4	1130,0
230	1130,6	1104,5	1088,4	1073,7	1090,9	1116,7	1105,2	1126,4	1106,7	1142,1	1096,6	1110,6
240	1114,3	1095,9	1082,0	1079,0	1098,3	1097,2	1090,0	1105,5	1077,6	1112,1	1090,4	1092,8
241	1105,7	1088,3	1077,8	1073,8	1094,3	1092,2	1087,6	1100,2	1078,0	1111,1	1085,7	1088,4

Layout of measuring points in the test furnace:

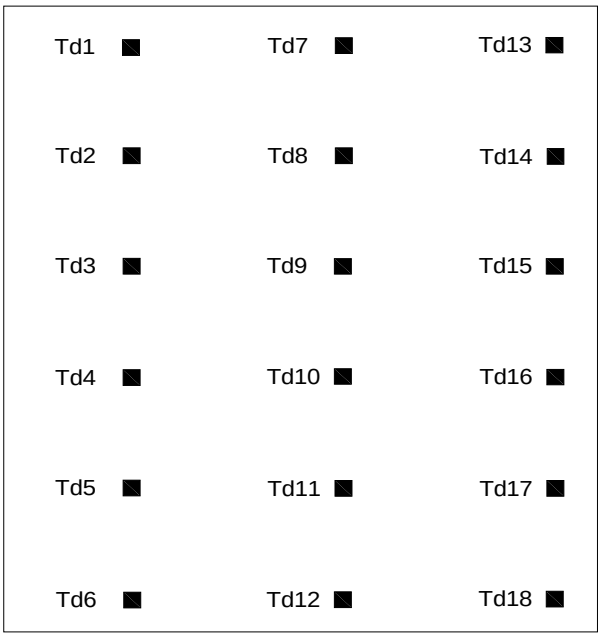


Measured values inside the test furnace – part 2

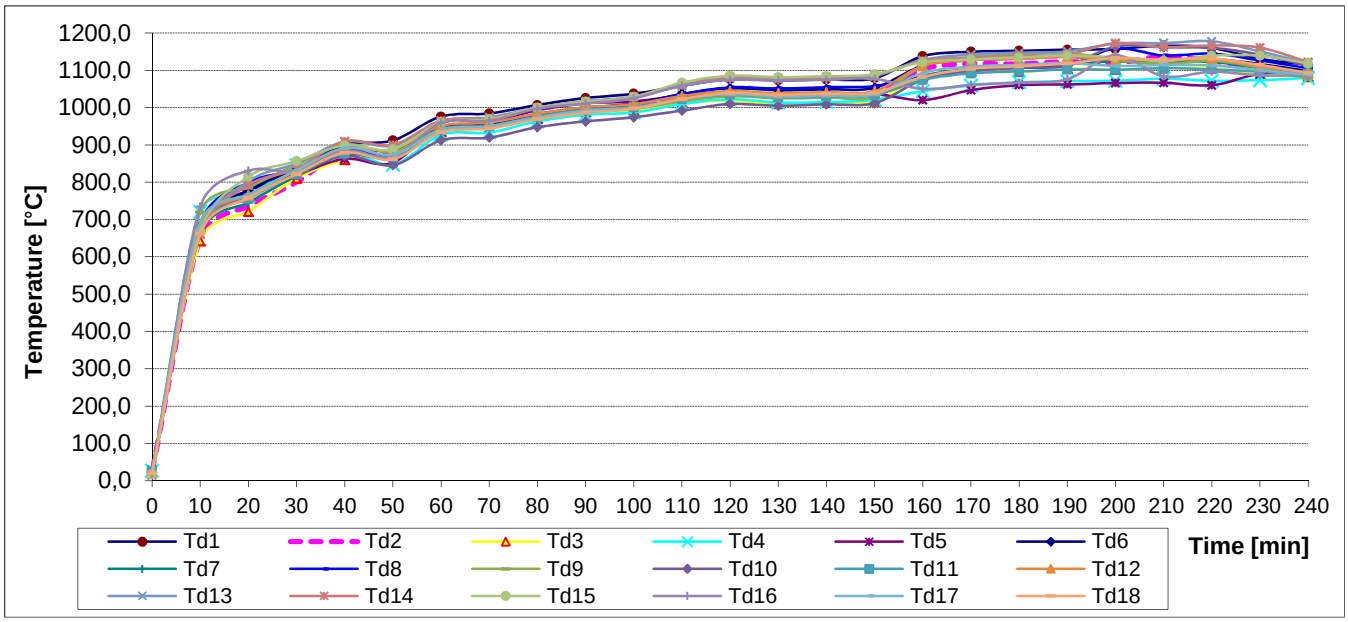
Time	Temperature [°C]									Deviation	Pressure [Pa]	
t [min]	Td13	Td14	Td15	Td16	Td17	Td18	Tave	Tn	To	d_e [%]	p1	p2
0	15,3	15,0	16,4	17,8	24,0	19,4	21,9	20,0	20,2	0,0	-0,5	-0,6
10	661,9	660,3	668,0	734,2	691,6	661,6	679,4	678,0	20,7	1,0	19,7	1,7
20	801,9	791,9	814,4	830,7	765,8	757,8	774,3	781,0	21,2	0,3	19,9	1,9
30	850,1	835,7	856,4	829,2	831,7	821,6	829,9	842,0	21,7	-0,1	17,3	-0,7
40	908,8	909,5	898,6	883,9	892,0	879,2	886,7	885,0	22,2	-0,5	19,9	1,9
50	897,8	900,2	888,1	869,4	874,0	861,8	875,9	918,0	22,7	-0,9	18,3	0,3
60	963,3	965,2	960,4	959,3	940,4	933,8	946,5	945,0	23,2	-0,7	18,8	0,8
70	973,9	971,1	968,6	964,9	949,6	943,8	955,3	968,0	23,7	-0,7	17,3	-0,7
80	999,8	999,9	999,8	997,1	974,0	968,9	981,3	988,0	24,2	-0,7	17,9	-0,1
90	1017,2	1016,2	1016,0	1010,6	991,7	986,2	998,6	1006,0	24,7	-0,7	19,6	1,6
100	1029,2	1028,8	1029,0	1025,0	1000,0	997,2	1008,2	1022,0	25,2	-0,8	18,2	0,2
110	1057,6	1062,5	1066,9	1059,1	1022,3	1023,3	1032,6	1036,0	25,7	-0,9	18,9	0,9
120	1078,4	1082,1	1085,5	1074,3	1036,0	1039,4	1047,4	1049,0	26,2	-0,8	18,1	0,1
130	1076,1	1078,4	1081,7	1072,8	1029,7	1033,9	1042,1	1061,0	26,7	-0,9	18,3	0,3
140	1080,1	1082,1	1084,0	1074,7	1032,1	1037,3	1044,8	1072,0	27,2	-1,0	18,8	0,8
150	1084,6	1087,8	1089,7	1076,7	1035,8	1039,9	1048,4	1082,0	27,7	-1,1	19,2	1,2
160	1127,8	1125,6	1121,8	1050,3	1092,9	1082,4	1091,2	1092,0	28,2	-1,0	19,9	1,9
170	1141,9	1133,3	1129,2	1061,1	1109,9	1104,0	1107,4	1101,0	28,7	-0,9	18,6	0,6
180	1146,8	1138,7	1134,7	1067,0	1114,7	1110,8	1112,8	1110,0	29,2	-0,8	17,3	-0,7
190	1151,2	1142,8	1139,3	1075,9	1119,4	1118,1	1118,1	1118,0	29,7	-0,8	19,7	1,7
200	1171,4	1173,1	1134,7	1140,3	1111,6	1133,1	1128,4	1126,0	30,2	-0,7	19,2	1,2
210	1172,9	1162,3	1124,8	1083,9	1114,5	1127,3	1124,4	1133,0	30,7	-0,7	18,3	0,3
220	1177,6	1166,4	1139,2	1096,3	1112,9	1131,3	1125,6	1140,0	31,2	-0,7	18,5	0,5
230	1150,5	1160,7	1138,8	1086,5	1100,4	1114,3	1113,5	1146,0	31,7	-0,8	17,5	-0,5
240	1119,7	1122,3	1118,1	1084,6	1087,4	1093,4	1097,8	1153,0	32,2	-0,9	19,3	1,3
241	1118,2	1121,6	1110,4	1071,3	1083,1	1093,2	1093,4	1153,0	32,2	-0,9	19,4	1,4

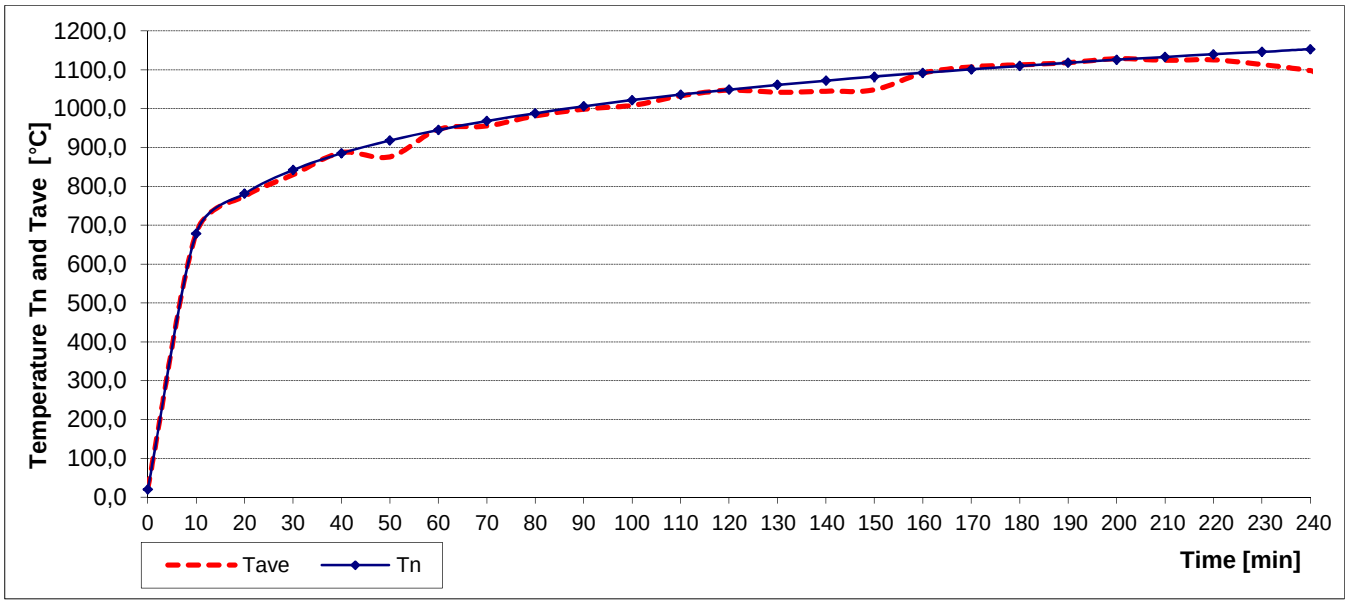
Tave Average temperature in the test furnace calculated from individual thermometers
Tn Standard temperature in the test furnace laid down according to test guideline
d_e Deviation of the average temperature from the standard temperature calc. acc. to test guideline
To Ambient temperature
p1 Pressure inside the test furnace on the top edge of specimen
p2 Pressure inside the test furnace at the height of neutral pressure level 500 mm above test furnace floor

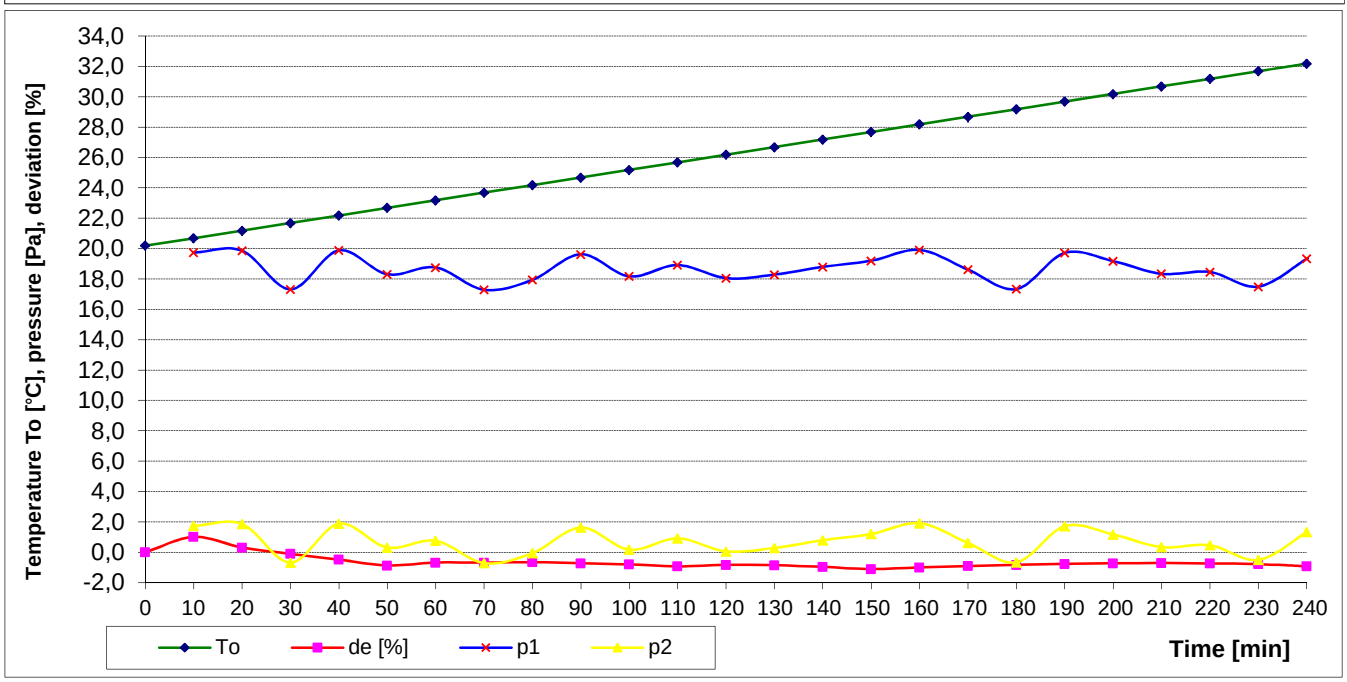
Layout of measuring points in the test furnace:



Measured values inside the test furnace /graph







Deflection of the specimen

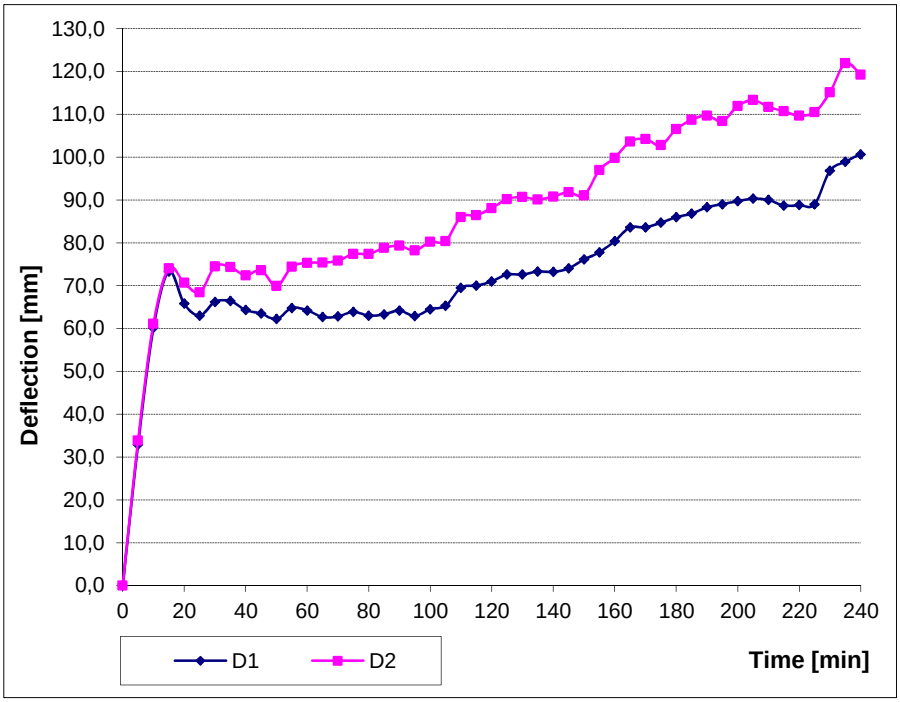
Time t [min]	Deflection [mm] D1	D2
0	0,0	0,0
5	32,9	33,8
10	60,1	61,1
15	73,3	74,0
20	65,8	70,7
25	63,0	68,4
30	66,2	74,5
35	66,4	74,3
40	64,3	72,4
45	63,5	73,6
50	62,2	69,9
55	64,8	74,4
60	64,2	75,3
65	62,7	75,4
70	62,8	75,8
75	63,9	77,4
80	63,0	77,4
85	63,3	78,8
90	64,2	79,3
95	62,9	78,2
100	64,5	80,2
105	65,3	80,4
110	69,5	86,0
115	70,0	86,4
120	71,0	88,1
125	72,6	90,2
130	72,6	90,7
135	73,3	90,1
140	73,2	90,8
145	74,0	91,8
150	76,1	91,1
155	77,8	97,0
160	80,4	99,8
165	83,6	103,6
170	83,6	104,2
175	84,7	102,8
180	86,0	106,5
185	86,8	108,7
190	88,3	109,7
195	89,0	108,4
200	89,7	111,9
205	90,3	113,3
210	90,0	111,7
215	88,7	110,7
220	88,8	109,7
225	89,0	110,5
230	96,8	115,1
235	98,9	121,9
240	100,6	119,2
241	99,6	119,0

Deflection [mm] deflection of the specimen surface

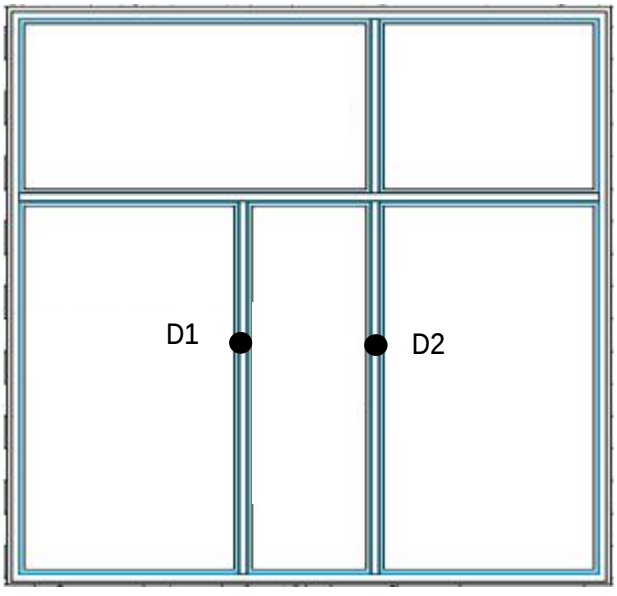
D1 - D2 measured with cable extension positioning transducer

Positive values of deflection represent deflection to the heat stress /
Negative values of deflection represent deflection from the heat stress.

Graph:



Layout of measuring points of deflection:



PHOTOS TAKEN DURING THE TEST



1st minute of the test



61st minute of the test



91st minute of the test

PHOTOS TAKEN DURING THE TEST



122nd minute of the test

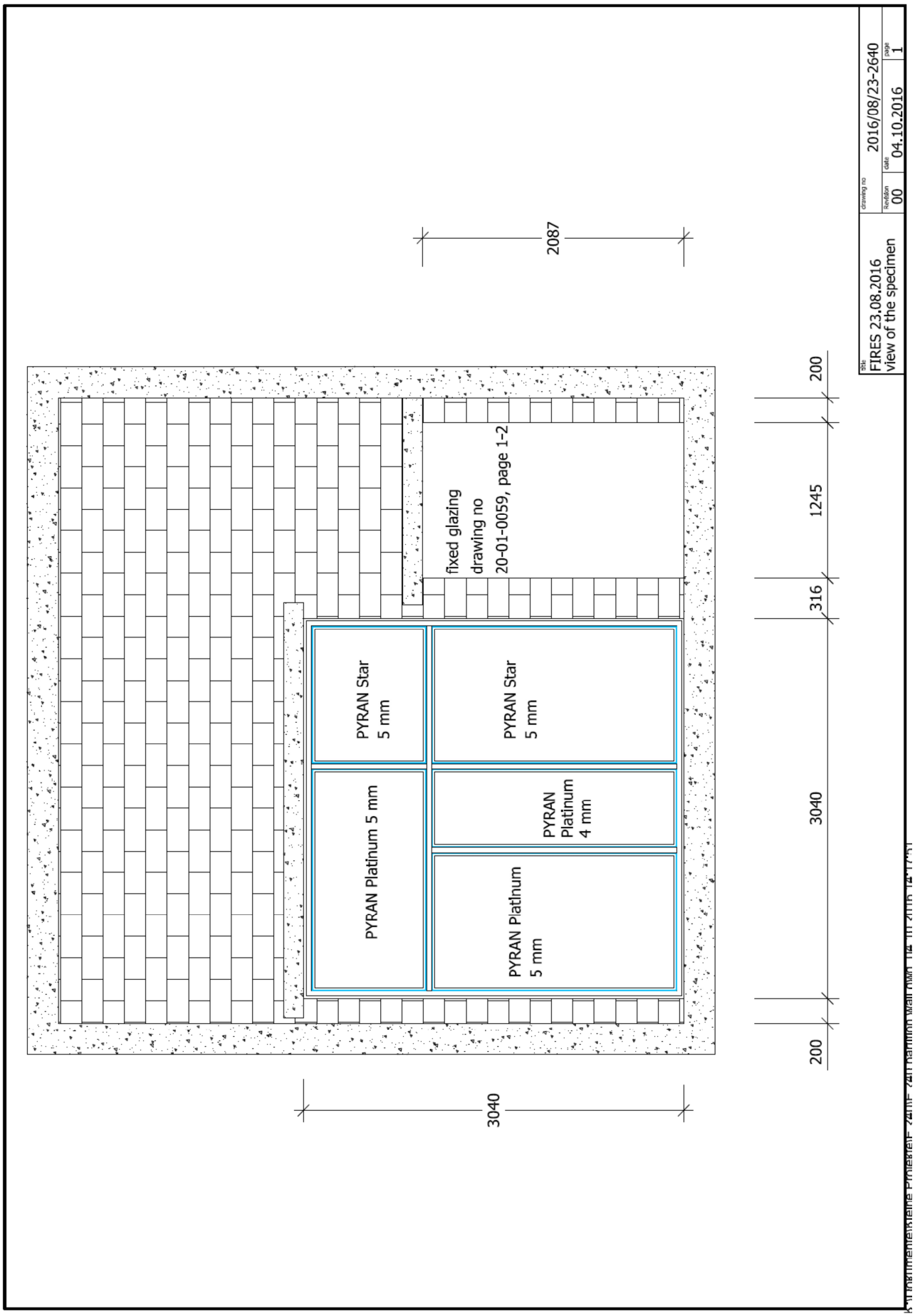


183rd minute of the test

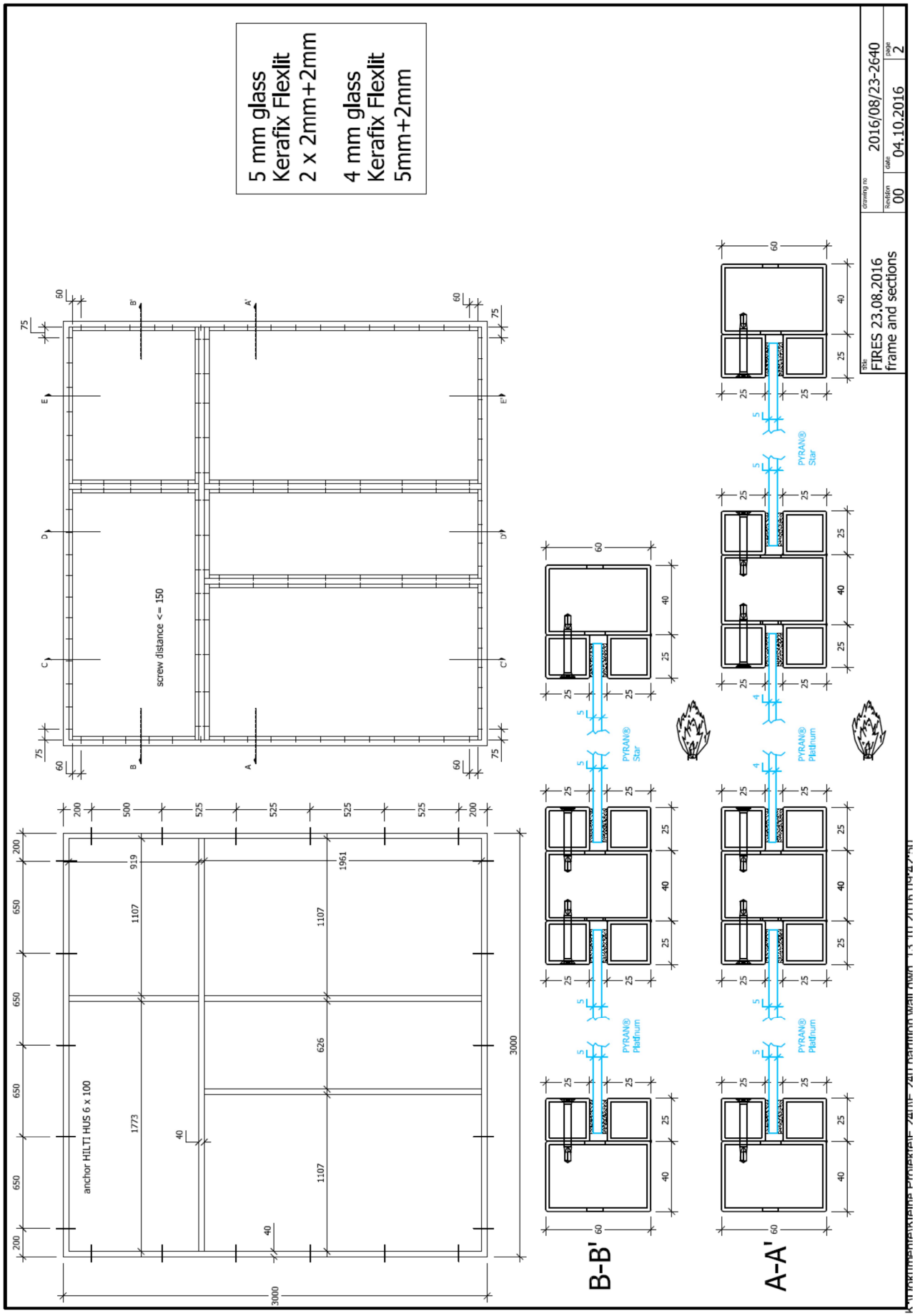


241st minute of the test

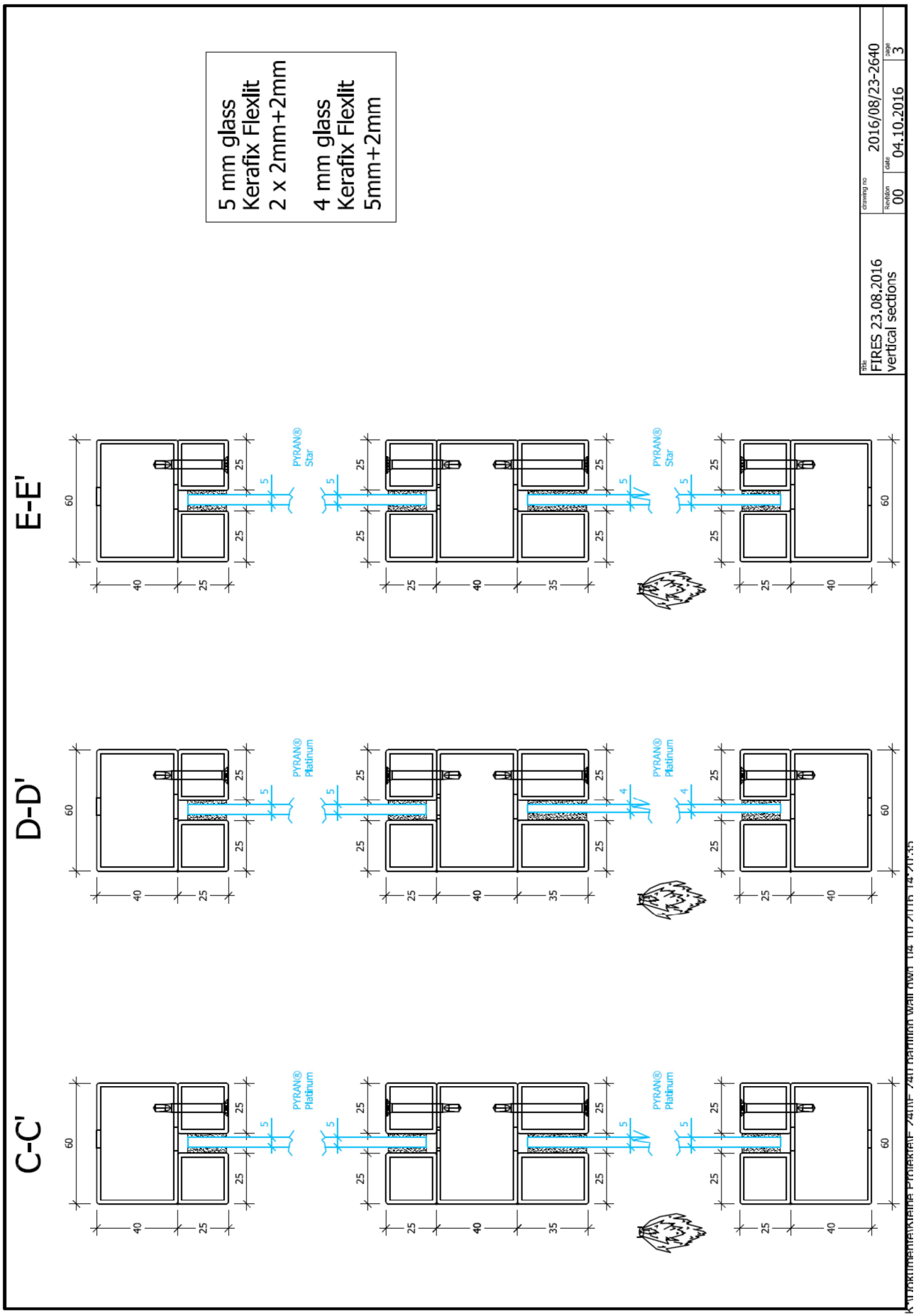
DRAWINGS


2087
200
1245
316
3040
200
3040
fixed glazing
drawing no
20-01-0059, page 1-2
PYRAN Star
5 mm
PYRAN Star
5 mm
PYRAN Platinum 5 mm
PYRAN
Platinum
4 mm
PYRAN Platinum
5 mm
title
FIRES 23.08.2016
view of the specimen
drawing no
2016/08/23-2640
Revision
00
date
04.10.2016
page
1

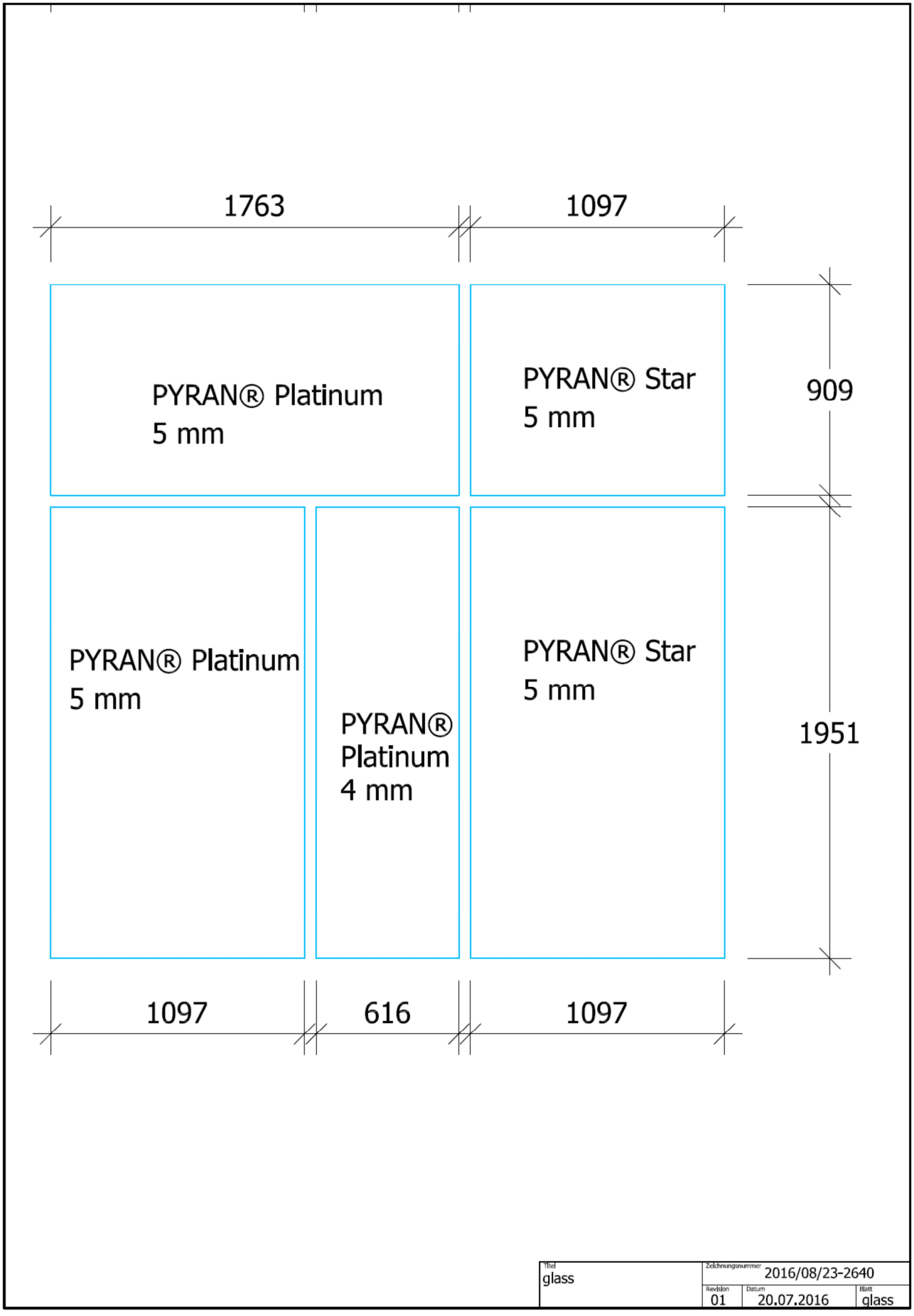
DRAWINGS


5 mm glass
Kerafix Flexlit
2 x 2mm+2mm
4 mm glass
Kerafix Flexlit
5mm+2mm
screw distance <= 150
anchor HILTI HUS 6 x 100
B-B'
A-A'
PYRAN® Star
PYRAN® Platinum
FIRES 23.08.2016
frame and sections
2016/08/23-2640
04.10.2016

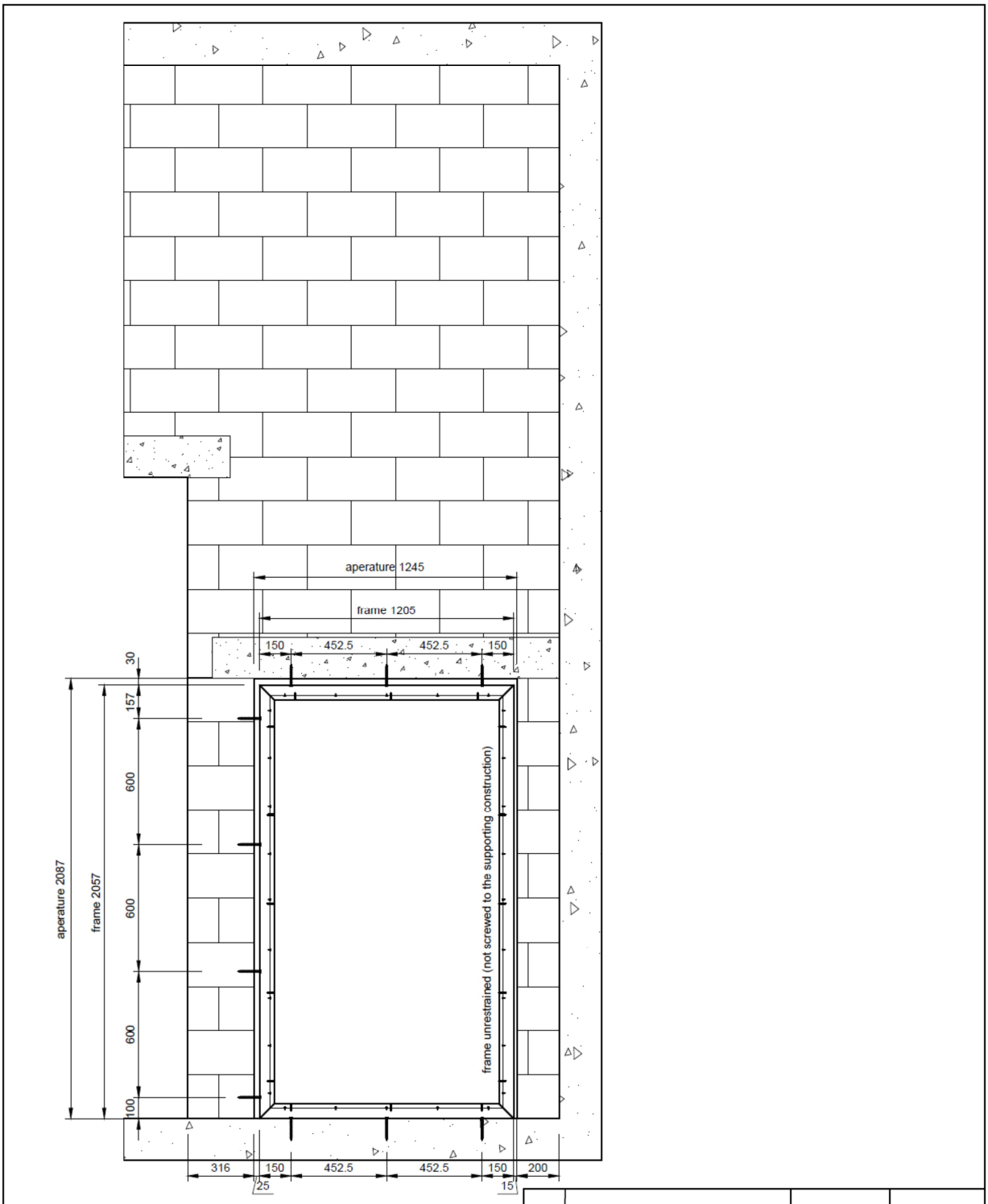
DRAWINGS


5 mm glass
Kerafix Flexlit
2 x 2mm+2mm
4 mm glass
Kerafix Flexlit
5mm+2mm
E-E'
D-D'
C-C'
PYRAN® Star
PYRAN® Platinum
FIRES 23.08.2016
vertical sections
2016/08/23-2640
04.10.2016
00
3

DRAWINGS


1763
1097
PYRAN® Platinum
5 mm
PYRAN® Star
5 mm
909
PYRAN® Platinum
5 mm
PYRAN®
Platinum
4 mm
PYRAN® Star
5 mm
1951
1097
616
1097
Titel
glass
Zeichnungsnummer
2016/08/23-2640
Revision
01
Datum
20.07.2016
Blatt
glass

DRAWINGS

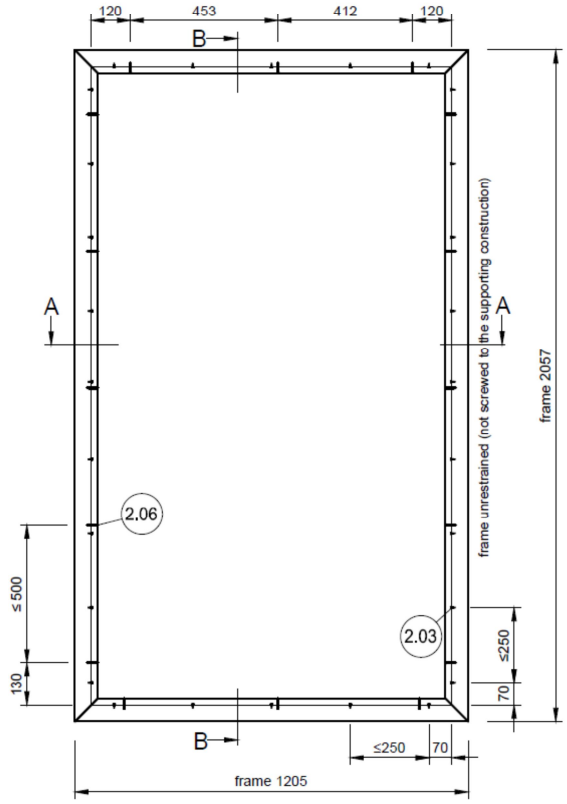


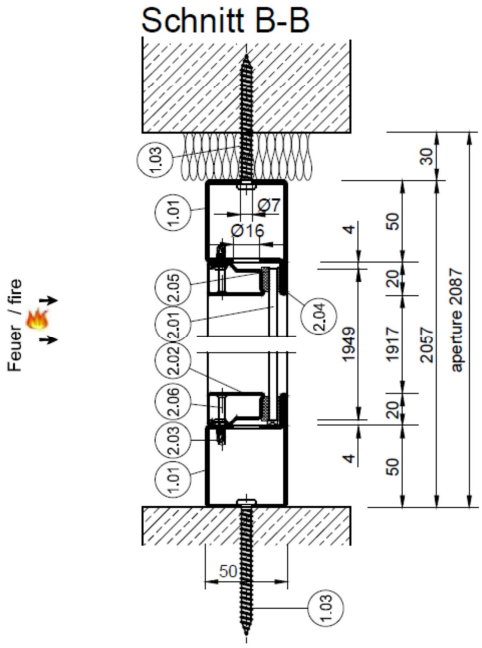
voestalpine Krems GmbH	Änderungs-Buchst.	Begründung	Tag / Name

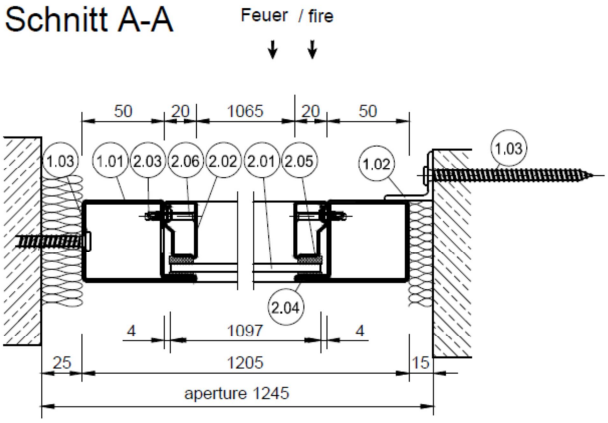
Die unbefugte bzw. bestimmungswidrige Verwendung dieser Unterlage ist verboten und würde gerichtlich verfolgt werden.

bearb. Name Datum	Münzberg 29.09.2016	Benennung	Zeichnungsnummer
geprüft Name Datum	Eckl 29.09.2016	fire test "vrame form 50" E 240 partition view - anchor	Format E; Nummer: 20-01-0059
			Blatt 1 von 2
Maßstab 1:25/1:4			Pfad: G:\Zng\20 RS, E 30, E 60, E 90, E 120\01\0059 FW_E 240, Schott-FIRES\20-01-0059 FW_E 240.dwg
			Speicherdatum: 10.10.2016

DRAWINGS







(1) frame

1.01	steel profile	"5050 L"
1.02	angular steel	"L30/30x3, L=2000mm "
1.03	Anchor	"Hilti HUS 6x100"

(2) glazing

2.01	Schott Platinium	"1097x1975x5mm"
2.02	glaszing bead	"GL 35-8"
2.03	stud	"BN65"
2.04	sealing strip	"EVB2x15"
2.05	sealing strip	"EVB2x15" (2x)
2.06	self drilling countersunk screw	"Ø3,9x32"

Änderg-Buchst.	Begründung	Tag	Name

voestalpine Krems GmbH

Die unbefugte bzw. bestimmungswidrige Verwendung dieser Unterlage ist verboten und würde gerichtlich verfolgt werden.

bearb. Name Datum	Münzberg	29.09.2016
geprüft Name Datum	Eckl	29.09.2016
Maßstab	1:20/1:4	

Benennung

fire test "vrame form 50" E 240

partition view, section, part list

Zeichnungsnummer

Format: E

Nummer: 20-01-0059

Blatt 2 von 2

Pfad: G:\Zng\20 RS, E 30, E 60, E 90, E 120\01\0059 FW_E 240, Schott-FIRES\20-01-0059 FW_E 240.dwg

Speicherdatum: 10.10.2016

8. FINAL PROVISION

§ This report details the method of construction, the test conditions and results obtained when the specific element of construction described herein was following the procedure outlined in EN 1363-1, and where appropriate EN 1363-2. Any significant deviation with respect to size, constructional details, loads, stresses, edge or end conditions other than those allowed under the field of direct application in the relevant test method is not covered by this report.

§ Because of the nature of the fire resistance testing and consequent difficulty in quantifying the uncertainty of measurement of fire resistance, it is not possible to provide a stated degree of accuracy of the result.

§ The test results refer only to the tested subjects. This test report is not an approval of the tested product by the test laboratory or the accreditation body overseeing the laboratory's activities. The test was carried out on testing equipment that is the property of FIRES, s.r.o., Batizovce. Without the written permission of the test laboratory this test report may be copied and/or distributed only as the whole. Any modifications of the test report can be made only by the fire resistance test laboratory FIRES, s.r.o., Batizovce.

Approved by:

Ing. Štefan Rástocký
leader of the testing laboratory

SKÚŠOBNÉ LABORATÓRIUM • PRÜFLABORATORIUM • TESTING LABORATORY • -1-
FIRES
The Experts on Fire Safety

Prepared by:

Bc. Dávid Šubert
technician of the testing laboratory

9. NORMATIVE REFERENCES

EN 1364-1: 2015	Fire resistance tests for non-loadbearing elements - Part 1: Walls
EN 1363-1: 2012	Fire resistance tests. Part 1: General requirements
EN 1363-2: 2001	Fire resistance tests. Part 2: Alternative and additional procedures

THE END OF THE TEST REPORT